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Epidemiology of Stingers in the National Football League, 2015-2019

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Background: Transient traumatic neuropraxia of either the brachial plexus or cervical nerve root(s) is commonly described as a "stinger" or "burner" by the athlete. Stingers in American Football commonly occur acutely as isolated injuries; however, concomitant injuries, including cervical spine pathologies, have also been reported.

Hypothesis: Among National Football League (NFL) athletes, the incidence rate of stingers is higher during the regular season than during the preseason and among positions with high velocity impacts such as running backs, linebackers, defensive backs, and receivers.

Study Design: Retrospective epidemiology study.

Level of Evidence: Level 4.

Methods: Aggregation of all in-game injuries with a clinical impression of "neck brachial plexus stretch" or "neck brachial plexus compression" entered into the NFL injury surveillance database through the centralized league-wide electronic medical record system over 5 years (2015-2019 seasons). Incidence rates per player-play were calculated and reported.

Results: A total of 691 in-game stingers occurred during the study period, with a mean of 138.2 per year. Average single-season injury risk for incident stinger was 3.74% (95% CI, 3.46%-4.05%). The incidence rate was higher during regular season games than during preseason games (12.26 per 100,000 player-plays [11.30-13.31] vs 8.87 [7.31-10.76], $P < 0.01$, respectively). The highest reported stinger incidence rates were among running backs and linebackers (both >15 per 100,000 player-plays). Among stingers, 76.41% did not miss time. Of those that resulted in time lost from football activities, mean time missed due to injury was 4.79 days (range, 3.17-6.41 days). Concomitant injuries were relatively low (7.09%).

Conclusion: In-game stinger incidence was stable across the study period and occurred most frequently in running backs and linebackers. Stingers were more common during the regular season, and most players did not miss time. Concomitant injuries were relatively rare.

Clinical Relevance: An improved understanding of the expected time loss due to stinger and concomitant injuries may provide insight for medical personnel in managing these injuries.

Keywords: brachial plexopathy; burner; cervical; National Football League; professional athletes; stinger

A forceful direct blow or traction injury to the neck or upper extremity can result in injury to the cervical nerve roots or upper trunk of the brachial plexus (C5-C7). These injuries are referred to as stingers or burners and occur commonly in collision sports including American football.[1,12,22,23,27] Stingers are the most common cervical spine injury and upper extremity nerve injury among elite American football players, accounting for approximately half of all cervical injuries.[26] Athletes often describe a burning pain that may radiate to the shoulder or arm and can be associated with

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weakness, numbness, and/or paresthesias.[4,15] Manual strength testing may present with transient weakness in the posterosuperior rotator cuff, deltoid, and biceps, which are innervated by the upper trunk.[2] Stingers occur from 1 of 3 mechanisms: (1) brachial plexus stretch/traction, (2) nerve root compression in the neural foramina, and (3) direct impact to the brachial plexus at Erb's point (between the shoulder pad and superomedial scapula).[1,17] A direct compression mechanism has been reported as particularly common among professional football players who are trained to specifically initiate blows through the shoulder.[7] An increased prevalence of stingers, particularly resulting from nerve root compression, has been reported in athletes with cervical and/or neural foraminal stenosis.[13,15,19,20] Because the cross-sectional area of the C4/5 neuroforamina is markedly smaller than the levels below, it is most commonly involved in stingers.[14] Symptoms are generally transient and resolve within seconds to minutes. Athletes may be returned to play only upon complete resolution of symptoms, which must include restoration of full painless neck motion and extremity strength.[1]

Previous studies have reported a high incidence of stingers as well as injury recurrence in American football players at various levels of competition. A study performed at the National Football League (NFL) Combine found that 17.4% of athletes had a history of cervical spine injuries, of which 62.2% were stingers.[5] Among collegiate football players, career stinger incidence has been reported as 49% to 65%, with recurrence rates up to 87%.[4,15] Previous studies have also reported a higher incidence at specific positions including running back, defensive linemen, linebackers, and defensive secondary.[24] Approximately 75% of football players who were diagnosed with a stinger were reported to miss no additional games due to injury.[1] Although these injuries have a relatively high incidence, stingers in football have been historically underreported due to the injury's generally transient nature, concerns about playing time, and long-term career implications.[1,6] As such, it has been difficult to determine the actual incidence of these injuries in football players at all levels of competition.

The purpose of this report was to describe the epidemiology of in-game stingers in the NFL from 2015 to 2019. Secondary objectives were to (1) examine the variation in incidence and missed time according to player position, (2) examine the variation in missed time according to number or previous stingers, and (3) describe the incidence of concomitant injuries.

METHODS

Study Design and Population

The current report retrospectively analyzed all recorded stingers occurring during all games (preseason, regular season, and postseason) of the 2015 through 2019 NFL seasons. Stingers that occurred during regular season and preseason practices, the offseason, or any other nongame activities were excluded from the study. All NFL players who played ≥1 NFL game during the study period were included.

Injury data were collected prospectively in a standardized manner for all 32 clubs as mandated by the NFL.[10] Starting in 2015, the NFL began using a centralized electronic health record (EHR) system to track all injuries to its athletes, regardless of time missed due to injury.[3]

In the NFL EHR, injuries with a clinical impression code of "Neck Brachial Plexus Stretch" or "Neck Brachial Plexus Compression" were considered stingers and were included in the study. Injury data were integrated with additional data from the NFL Game Statistical and Information System (GSIS), which records player participation in games (eg, number of plays).[3] Missed time was defined as the number of days elapsing between the date of injury and the date of return to full participation in football activities as reported by the medical staff.[3]

Data collected included player position at the time of injury (offensive line, running back, tight end, quarterback, wide receiver, defensive secondary, defensive line, linebacker, kicker/punter/long snapper), injury timing (preseason, regular season, postseason game), time missed due to injury, concomitant injuries, laterality of injury, primary mechanism of injury, number of previous stingers while in the NFL, player height, and player weight. Position at time of injury combines all players playing on special teams plays into 1 category, whereas roster position only includes kickers, punters, and long snappers in the special teams category.

The study received Emory University Institutional Review Board (IRB) approval (IRB No. 00002783) as well as approval from the NFL Player Scientific and Medical Research Approval Process.[16]

Analytic Methods

Descriptive statistics were calculated, including counts, percentages, and means. To account for the number of unique players participating in football activities each year, incidence was calculated as a 1-season stinger risk both by season and as a 5-season average, along with 95% CIs calculated using the exact binomial distribution. When determining 1-season game-related stinger risk, athletes with multiple stingers in the same year were counted only once per year. Incidence rate of game-related stingers was calculated as the rate per 100,000 player-plays to account for the number of plays in which each individual player participated as recorded in the GSIS. A single exposure, or player-play, was defined as 1 player participating in a single offensive, defensive, or special teams play. Player-games were defined as the number of unique players participating in a single game, regardless of the duration of their participation. Concomitant injuries were categorized as concussion, cervical spine injury (including diagnoses of cervical disc derangement, cervical disc herniation, cervical fracture, cervical stenosis, and cervical strain), shoulder injury (including diagnoses of shoulder contusion, shoulder strain, acromioclavicular joint sprain, glenoid labral tear, shoulder nerve disorder, rotator cuff tear, deltoid tear), or other (including diagnoses of chest/rib fracture, sternoclavicular joint

Table 1. Stingers in the NFL, 2015-2019[a]

	2015	2016	2017	2018	2019	5-Year Average	Total
Stingers	124	128	136	163	140	138.2	691
Season							
Preseason	13	20	20	24	26	20.6	103
Regular season	108	106	115	131	111	114.2	571
Postseason	3	2	1	8	3	3.4	17
Injury mechanism							
Direct impact	80	74	88	106	84	86.4	432
Impingement	1	2	4	11	3	4.2	21
Indirect force	12	19	11	0	0	8.4	42
Shearing	3	0	1	0	1	1	5
Spontaneous/Insidious	2	2	0	1	0	1	5
Stretch	22	23	23	27	30	25	125
Torsion	1	1	4	1	1	1.6	8
External rotation	0	0	0	1	0	0.2	1
Hyperextension	0	0	0	2	7	1.8	9
Unknown/inconclusive	3	7	5	14	14	8.6	43
Unknown/missing	0	0	0	1	1	0.4	2
Player demographics[b]							
Weight, lb (average)	246.74	250.52	249.33	244.01	246.79	247.48	n/a
Height, in. (average)	74.02	74.27	74.18	73.66	74.09	74.044	n/a

NFL, National Football League.
[a]Values are presented as number (n).
[b]Weight and height reflect values recorded at the time when the player joined the club where he was injured.

sprain, rhomboid strain, trapezius strain). Incidence rates, incidence rate ratios (IRRs), and 95% CIs were calculated using the Poisson distribution. Poisson regression was used to generate *P* values, with statistical significance accepted for $P < 0.05$.

RESULTS

A total of 691 game-related stingers occurred during the 2015 to 2019 NFL seasons (Table 1, Figure 1), with an average of 138.2 stingers per year and a relatively stable incidence (range, 124-163) across the study period. There were 507 (73.4%) brachial plexus stretch injuries and 184 (26.6%) brachial plexus compression injuries.

During the study period, the average single-season game-related stinger risk was 3.74% (95% CI, 3.46%-4.05%; Table 2).

There was a higher incidence rate of stingers in the regular season compared with the preseason (12.26 per 100,000 player-plays [95% CI, 11.30-13.31] vs 8.87 per 100,000 player-plays [95% CI, 7.31-10.76], $P < 0.01$) (Table 3).

For all players, the overall incidence of stingers in games was 11.48 per 100,000 player-plays (95% CI, 10.66-12.37), with an increased incidence among running backs (18.00, [95% CI, 14.03-23.09]), linebackers (15.87, [95% CI, 13.48-18.68], and defensive linemen (14.78, [95% CI, 12.39-17.64]) when compared with the average (Table 4).

Overall, 76.41% of athletes who sustained a stinger missed no additional time due to injury, and the average time missed due

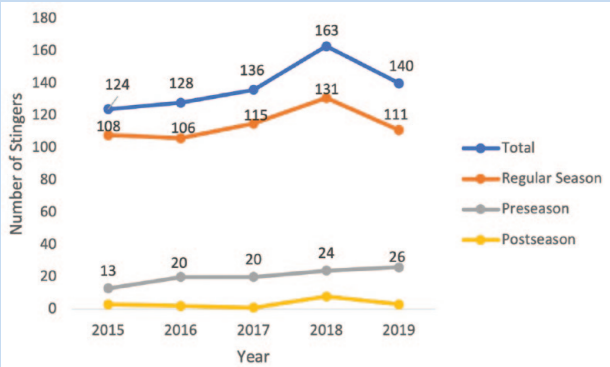

Figure 1. Number of NFL game-related stingers each year from 2015 to 2019 in the regular season, preseason, and postseason. NFL, National Football League.

to injury was 4.79 days (95% CI, 3.17-6.41). More than half of athletes at each position missed no additional time due to injury (Table 5). For brachial plexus stretch injuries, the average time missed due to injury was 4.20 days (95% CI, 1.76-5.96), with a maximum time loss of 236 days. For brachial plexus compression injuries, the average time missed due to injury was 5.98 days (95% CI, 2.42-9.55), with a maximum time loss of 280 days.

When stratifying according to the number of previous stingers (0, 1, 2, or more), there was also no difference in the proportion of athletes missing additional time due to injury or in the number of days missed due to injury (Table 6).

Concomitant injuries were relatively uncommon (n = 45, 6.51%; Figure 2). The most frequent concomitant injuries were concussions (n = 10, 22.22% of concomitant injuries), shoulder injuries (n = 18, 40.00% of concomitant injuries), and cervical spine injuries (n = 14, 31.11% of concomitant injuries). With respect to cervical spine injuries, there was "neck cervical disc herniation C3-C4" in 1 patient, "neck cervical disc herniation C4-5" in 1 patient, "neck cervical disc herniation C5-6" in 3 patients, "neck cervical disc herniation C6-C7" in 2 patients, "neck cervical disc derangement" in 1 patient, "neck cervical disc disease" in 1 patient, "neck cervical spinal stenosis" in 4 patients, and "neck cervical fracture C5-C6" in 1 patient.

DISCUSSION

In this descriptive epidemiological study of game-related stingers in the NFL, we report a relatively stable injury incidence across the study period, a relatively higher injury incidence among running backs and linebackers, and a relatively low incidence of concomitant injuries. In addition, most athletes missed no additional time due to injury after the game in which the injury occurred, regardless of whether a primary or recurrent injury. To our knowledge, this is the first epidemiologic report of game-related stingers in the NFL. An improved understanding of the expected time loss due to stinger and concomitant injuries may provide insight for medical personnel in managing these injuries.

Previous studies have reported a high incidence of stingers in collision sports, including American football and rugby.[6,22] In a 2014 study of football athletes at various levels (high school, collegiate, and professional), 50.3% of players reported a history of at least 1 stinger during their playing career.[24] Another study of collegiate athletes found that 23% of incoming National Collegiate Athletic Association (NCAA) Division I football players reported a history of a stinger, and over 50% of existing Division I football athletes reported a history of multiple stingers.[21] Increasing years played and a history of stingers have been associated with stinger injury recurrence.[8] Another study of NCAA football athletes demonstrated that stingers were 4 times more likely to occur in games than in practice.[9] In the present study, we found a higher injury incidence rate during regular season games when compared with preseason games. Meanwhile, a previous study that investigated stinger incidence in NCAA football players reported an increased incidence during preseason practice when compared with in-season practice. [25] The authors purported that this observed increase in preseason stinger incidence could be attributable to a lower fitness level during the preseason as well as athletes' unwillingness to report injuries during the regular season due to concerns of lost playing time. As these athlete populations differ in terms of age, level of competition, and preseason activities, there are several possible explanations for these findings. First, there is a markedly increased level of competition during regular season NFL games compared with the preseason, which may increase the risk of stingers. It is also possible that stingers are relatively underreported by NFL players during the preseason because of nonmedical reasons. Previous studies have suggested that athletes are prone to underreport stingers due to potential career implications.[6,18] We found that greater than three-quarters of athletes missed no additional days due to injury - a figure consistent with previous studies on stingers in football players.[6]

When analyzing injury incidence by position, we found that running backs and linebackers had the highest incidence of injury. This finding was not unexpected, as these positions engage in high velocity collisions over variable distances more frequently than other positions. While offensive and defensive linemen collide on nearly every play, the velocity at which these collisions occur is generally lower than running backs and linebackers. Furthermore, the techniques used for blocking or evading a block (used by offensive and defensive linemen, respectively) differ markedly from the collisions experienced by running backs and linebackers. While high velocity collisions do occur at several other positions, including defensive back and wide receiver, these players have many plays in which no collisions occur at all. Therefore, their collision incidence per player-play is lower, thus likely contributing to their lower stinger incidence. A previous epidemiological study that investigated the incidence of injuries resulting in shoulder instability among NFL athletes similarly found a high incidence of injury at the running back and linebacker positions.[3] As the

Table 2. One-Season Game-Related Stinger Risk

Year	Game Stingers, Number of Players	One-Season Game Stinger Risk, %	95% CI[a]
2015	111	3.39	2.79, 4.06
2016	111	3.38	2.79, 4.05
2017	120	3.60	2.99, 4.28
2018	143	4.29	3.63, 5.03
2019	129	4.07	3.41, 4.82
5-year average	122.8	3.74	3.46, 4.05

[a]Exact binomial 95% CIs. Athletes with multiple stingers in the same year were only counted once per year.

Table 3. Incidence of Game-Related Stingers by Season From 2015 to 2019

Season	Game Injuries	Player-Plays	Rate[a]	95% CI[b]
Total	691	6,018,765	11.48	10.66, 12.37
Preseason	103	1,161,463	8.87	7.31, 10.76
Regular season	571	4,657,221	12.26	11.30, 13.31
Postseason	17	200,081	8.50	5.28, 13.67

[a]Rate is per 100,000 player-plays.
[b]Poisson 95% CIs.

Table 4. Incidence of Game-Related Stingers by Roster Position From 2015 to 2019

	Stingers, n	Player-Plays	Rate[a]	95% CI[b]
Total	691	6,018,758	11.48	10.66, 12.37
Offensive line	99	1,224,298	8.09	6.64, 9.85
Running back	62	344,469	18.00	14.03, 23.09
Tight end	45	360,059	12.50	9.33, 16.74
Quarterback	6	229,910	2.61	1.17, 5.81
Wide receiver	23	661,839	3.48	2.31, 5.23
Defensive secondary	188	1,368,232	13.74	11.91, 15.85
Defensive line	123	832,030	14.78	12.39, 17.64
Linebacker	144	907,508	15.87	13.48, 18.68
Special teams[c]	1	90,413	1.11	0.16, 7.85

[a]Rate is per 100,000 player-plays.
[b]Poisson 95% CIs.
[c]Special teams roster position only includes kickers, punters, and long snappers.

Table 5. Missed Days Based on Position at Time of Injury

Player Position	Injuries, n	No Missed Days, %	Mean Days Missed	Standard Deviation
Offensive line	97	66.67	8.64	20.0
Running back	48	79.17	4.15	15.1
Tight end	40	85.00	1.05	3.0
Quarterback	7	57.14	3.43	5.7
Wide receiver	14	64.29	5.54	8.8
Defensive secondary	167	80.24	5.07	24.2
Defensive line	117	76.92	3.29	14.9
Linebacker	106	75.47	3.57	19.7
Special teams[a]	95	77.90	5.71	24.7

[a]Special teams at time of injury includes all players playing on special teams.

Table 6. Missed Time Based on Number of Previous Stingers

No. of Previous Stingers	Injuries, n (%)	No Missed Days, %	Mean Days Missed	Standard Deviation
0	450 (65.12)	76.89	3.89	18.3
1	156 (22.58)	75.64	6.74	23.8
2 or more	85 (12.30)	75.29	5.86	24.6

mechanisms of injury and energy required to cause shoulder instability events and stingers are similar, these findings are not unexpected. In fact, shoulder injuries comprised among the highest proportion of concomitant injuries in our current study.

Although there was a relatively low incidence of concomitant injuries that occurred with stingers in this study (approximately 7%), concussion was a common concomitant injury sustained. Currently, the NFL concussion protocol mandates that the on-field portion of the concussion evaluation is performed when an athlete is diagnosed with a stinger.[11] Concussions cannot be missed due to the implications of repeated head trauma and its associated long-term health implications. In addition to concussion assessment, a standard evaluation after a stinger includes a full motor and sensory evaluation to ensure restoration to baseline levels before permitting the athlete to return to participation. This standard evaluation will identify most acute cervical spine injuries, but there is still a required follow-up period postgame and in the subsequent days after the game in which additional imaging and treatment plans ensure that other concomitant injuries are diagnosed and treated accordingly. Mack et al[16] conducted a study on concussions in the NFL over the same period as our study and reported a 5-year concussion rate of 17.0 concussions per 100,000 player-plays. Analysis by position showed that concussion incidence was highest in defensive secondary (cornerbacks, safeties, and defensive backs) and offensive linemen.[16] Reported game concussion incidence decreased by 23% in the last 2 years of the 5-year study period from 2015 to 2019.[16] In the present study, total game-related stinger incidence spiked during the 2018 season before decreasing in the final year of this 5-year study. Before the start of the 2018 season, the NFL instituted a rule making it illegal for a player to lower his head to initiate contact with his helmet against an opponent. When this rule was instituted, the NFL provided an "NFL Way to Play" educational series along with the memo to athletic trainers, providing specific guidelines on tackling techniques. The observed reduction in concussions at the end of the study period in the present study and that of Mack et al[16] may be due in part to the addition of head and neck injury reduction efforts and education, detailed injury reviews, and rule changes targeting high-risk play types.[1,16] However, a corresponding decrease in stinger incidence has not been observed, and in fact

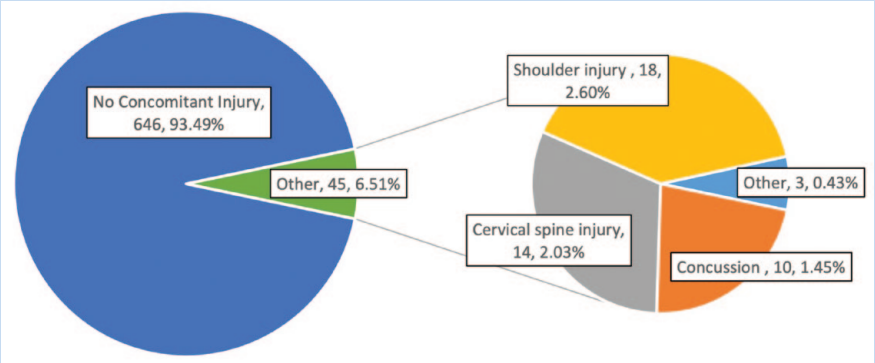

Figure 2. Concomitant injuries to NFL players who experienced in-game stingers during the 2015 to 2019 seasons. "Cervical spine injury" includes diagnoses of cervical disc derangement, cervical disc disease, cervical disc herniation, cervical stenosis, and cervical fracture. "Shoulder injury" includes diagnoses of shoulder contusion, shoulder strain, shoulder sprain, acromioclavicular sprain, glenoid labral tear, shoulder nerve disorder, and rotator cuff tear. "Other" includes diagnoses of elbow contusion, rhomboid strain, sternoclavicular sprain, and chest fracture/rib injury. NFL, National Football League.

2018 had the highest incidence any year during the study period. Future investigations will track stinger injury rates over the next decade to ensure that current legal tackling techniques are not increasing the incidence of stingers.

Limitations

The findings of this study may not be generalizable to American football athletes at lower levels of play such as collegiate, high school, and youth. Although utilizing the current NFL injury database minimized the risk of reporting inaccuracies, injuries that were not reported by players or the clubs were not captured. It is likely that stingers continue to be underreported among symptomatic athletes, and only those that were diagnosed and documented in the NFL EHR were captured in this study. Furthermore, the number of individual plays or series missed due to stinger injury during the game in which the injury occurred was not available. Player age was not available and, as such, we were unable to determine whether there was an association between age and mechanism of stinger injury (brachial plexus compression versus stretch injury). In the NFL injury severity scores (ISS), no specific clinical impression code for cervical nerve root compression or transient cervical neurapraxia or exists, and stinger injuries are typically recorded by each team's medical staff as either "neck brachial plexus compression" or "neck brachial plexus stretch" after a thorough history, physical examination, and video review of the play or activity in which the injury occurred. As such, while nerve root compression in the neural foramina has been described as a distinct mechanism for stingers, we were unable to determine which injuries may have fallen into this category, and it is also possible that some of these injuries may have been missed.[1,17] Even when video review is performed, it may be difficult to accurately categorize an injury as either brachial plexus stretch or compression, introducing a potential for the inaccuracy in coding the mechanism of injury in the ISS. In addition, imaging studies and reports, such as magnetic resonance imaging were not available for review. We were also unable to assess whether any injuries resulted in permanent neurologic deficits, as the only outcome data available were time missed due to injury. To our knowledge, all athletes did return to sport. Finally, injury recurrence information was available only while players were participating in the NFL. Injuries that occurred before NFL participation or while the player was not on an NFL club roster were not included.

CONCLUSION

This is the first epidemiologic report of game-related stinger injuries in NFL athletes. Stinger incidence was relatively stable during the study period and occurred most frequently in running backs and linebackers. Injuries were more common during regular season games, and most players missed no additional days beyond the game in which the injury occurred. While concomitant injuries were relatively uncommon, injuries including concussions and cervical spine injuries did occur and must be ruled out before returning an athlete to participation. An improved understanding of the expected time loss due to stinger and concomitant injuries may provide insight for medical personnel in managing these injuries.

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